

Week 3

$$\begin{cases} x + 2y - 3z = 4 \\ 3x - y + z = 5 \\ x + 3y + 0z = 1 \end{cases}$$

m - rows

$$\begin{cases} a_{11} & a_{12} & a_{13} & \dots & b_1 \\ a_{21} & a_{22} & a_{23} & \dots & b_2 \\ \vdots & \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \dots & b_m \end{cases}$$

Augmented matrix

number of equations

Ex

$$n = 3 - \text{rows} \left\{ \begin{array}{cccc} \textcircled{1} & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{array} \right.$$

$(n+1) = (3+1) = 4 \text{ columns}$

augmented matrix

↳ play with augmented matrix

↳ get different matrix

↳ a different S.L.E

will have some solution to initial $\Rightarrow$ But much simpler

Aug $\Rightarrow$ S.L.E

$$\begin{bmatrix} 2 & 1 & 5 \\ 6 & 7 & 9 \end{bmatrix}$$

$\xrightarrow{?}$

of rows $m = 2$

of columns $n+1 = 3$

$\Rightarrow n = 2$

of variables

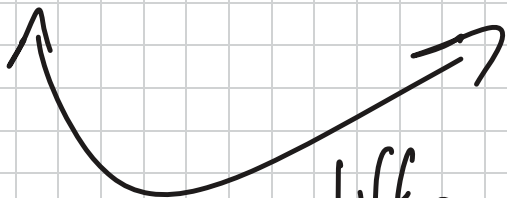
(x_1, x_2)

$$2x_1 + 1x_2 = 5$$

$$6x_1 + 7x_2 = 9$$

$$\begin{cases} 1 & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{cases}$$

$$\begin{cases} 1 & 3 & 0 & 1 \\ 3 & -1 & 1 & 5 \\ 1 & 2 & -3 & 4 \end{cases}$$



different SLE
 same solution
 "elementary operations"

$$3 \quad -1 \quad 1 \quad 5 \quad \Rightarrow \quad (-3) [3 \quad -1 \quad 1 \quad 5]$$

Some solution

$$-9 \quad 3 \quad -3 \quad -15$$

$$\begin{array}{l} \text{LHS}_1 = \text{RHS}_1 \\ \text{LHS}_2 = \text{RHS}_2 \end{array} \quad \rightarrow \quad \text{LHS}_1 + \text{LHS}_2 = \text{RHS}_1 + \text{RHS}_2$$

$$\text{LHS}_1 + \# \text{LHS}_2 = \text{RHS}_1 + \# \text{RHS}_2$$

$$\text{LHS: } x + 2y - 3z + 3(3x - y + z)$$

$$= x + 2y - 3z + 9x - 3y + 3z$$

$$= 10x - y + 0 \cdot z$$

$$\text{RHS} = 19$$

Some solutions

$$\begin{Bmatrix} 2 & 1 & 5 \\ 6 & 7 & 9 \end{Bmatrix} \xrightarrow{r_1 \rightarrow r_1 - 2r_2} \begin{Bmatrix} -10 & -13 & -13 \\ 6 & 7 & 9 \end{Bmatrix}$$

$$\begin{aligned} (2 \ 1 \ 5) + (-2) \cdot (6 \ 7 \ 9) &= \\ = (2 \ 1 \ 5) + (-12 \ -14 \ -18) &= \\ = (-10, -13, -13) \end{aligned}$$

*reating an owl to Clicks

REF matrix

First non zero # is always 1
("leading one")

rows of 0
must be at
the bottom

$$\begin{matrix} r_1 \\ r_2 \\ r_3 \\ r_4 \end{matrix} \left\{ \begin{array}{cccccc} 0 & 1 & 2 & 6 & 0 & 1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right\}$$

Next row has to
Shift at least once

if all rules are met
it is in REF

RREF

↑
reduced

$$\begin{matrix} & C_1 & C_2 & C_3 & C_4 & C_5 & C_6 \\ \left[\begin{array}{cccccc} 0 & 1 & 2 & 6 & 9 & 1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right] \end{matrix}$$

or "0"

← matrix REF

BUT not RREF

Elementary row operations

$$\begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 0 & 3 \end{bmatrix} \leftarrow \begin{matrix} r_1 = [2 \ 1] \\ r_2 = [-1 \ 5] \\ r_3 = [0 \ 3] \end{matrix}$$

↪ interchange: Ex
 $r_1 \leftrightarrow r_3$

$$\begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 0 & 3 \\ -1 & 5 \\ 2 & 1 \end{bmatrix}$$

x by a #

Ex
 $r_2 \rightarrow r_2 \cdot 7$

$$\begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 0 & 3 \end{bmatrix} \rightarrow \begin{bmatrix} 2 & 1 \\ -7 & 35 \\ 0 & 3 \end{bmatrix}$$

$r_a \rightarrow r_a + \# r_b$

$r_3 \rightarrow r_3 - 2r_1$

$$\begin{bmatrix} 2 & 1 \\ -1 & 5 \\ 0 & 3 \end{bmatrix} \rightarrow$$

$\begin{bmatrix} 0 & 3 \\ 4 & 2 \end{bmatrix} \leftarrow 2r_1$

$$\begin{array}{c} \overline{[-4 \quad 1]} \\ \uparrow \quad \nwarrow \\ 0-4 \quad 3-2 \end{array}$$

$$A \rightarrow A_{REF} \rightarrow A_{REF}$$

Using 1 missed it

$$\begin{array}{c} \checkmark \quad \checkmark \quad \times \\ \downarrow \quad \downarrow \quad \downarrow \\ r_1 \quad r_2 \quad r_3 \quad r_4 \end{array} \begin{bmatrix} 0 & 1 & 2 & 6 & 0 & 1 \\ 0 & 0 & 1 & -1 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$r_1 \rightarrow r_1 - 2 \cdot r_2$
opposite to 2

$$\begin{array}{l} [0 \ 1 \ 2 \ 6 \ 0 \ 1] \\ [0 \ 0 \ 2 \ -2 \ 4 \ 0] \\ \hline [0 \ 1 \ 0 \ 4 \ -4 \ 1] \end{array}$$

$$\begin{array}{c}
 \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \times \\
 r_1 \left[\begin{array}{ccc|cc} 0 & 1 & 0 & 8 & -4 & 1 \end{array} \right] \\
 r_2 \left[\begin{array}{ccc|cc} 0 & 0 & 1 & -1 & 2 & 0 \end{array} \right] \\
 r_3 \left[\begin{array}{ccc|cc} 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right] \\
 r_4 \left[\begin{array}{ccc|cc} 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

$$r_1 \rightarrow r_1 - (-4)r_3$$

$$r_2 \rightarrow r_2 - 2r_3$$

$$r_1 \rightarrow \underbrace{r_1 + 4r_3}_{\text{new } r_1}$$

$$\begin{array}{r}
 0 \ 1 \ 0 \ 8 \ -4 \ 1 \\
 + \ 0 \ 0 \ 0 \ 0 \ 4 \ 4 \\
 \hline
 [0 \ 1 \ 0 \ 8 \ 0 \ 5]
 \end{array}$$

how do
we know
what's red?

$$\begin{array}{c}
 \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \quad \checkmark \\
 r_1 \left[\begin{array}{ccc|cc} 0 & 1 & 0 & 8 & 0 & 5 \end{array} \right] \\
 r_2 \left[\begin{array}{ccc|cc} 0 & 0 & 1 & -1 & 0 & -2 \end{array} \right] \\
 r_3 \left[\begin{array}{ccc|cc} 0 & 0 & 0 & 0 & 1 & 1 \end{array} \right] \\
 r_4 \left[\begin{array}{ccc|cc} 0 & 0 & 0 & 0 & 0 & 0 \end{array} \right]
 \end{array}$$

$$r_2 \rightarrow r_2 - 2r_3$$

} RREF

$$x + 2y - 3z = 4$$

$$3x - y + z = 5$$

$$x + 3y + 0z = 1$$

$$\Rightarrow A = \begin{bmatrix} 1 & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

leading one (pointing to the first element of r_1)

Helper row (pointing to r_1)

must be zero (pointing to the first element of r_3)

$$r_2 \rightarrow r_2 + (-3) \cdot r_1$$

and

$$r_3 \rightarrow r_3 + (-1) \cdot r_1$$

$$\rightarrow \begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & -7 & 10 & -7 \\ 0 & 1 & 3 & -3 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

✓ x (pointing to the first element of r_3)

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & -7 & 10 & -7 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

✓ ✓ (pointing to the first and second elements of r_3)

$$r_3 \rightarrow \begin{bmatrix} 1 & 3 & 0 & 1 \end{bmatrix}$$

$$+ \begin{bmatrix} -1 & -2 & 3 & -4 \end{bmatrix}$$

$$\begin{bmatrix} 0 & 1 & 3 & -3 \end{bmatrix}$$

new r_3 (pointing to the new r_3)

$$r_3 \rightarrow r_3 + 7 \cdot r_2$$

Don't touch

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 31 & -28 \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

✓ ✓

$$r_3 \rightarrow \frac{r_3}{31}$$

REF

How does it help us?

$$\begin{bmatrix} 1 & 2 & -3 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 1 & -\frac{28}{31} \end{bmatrix} \begin{matrix} r_1 \\ r_2 \\ r_3 \end{matrix}$$

✓ ✓ ✓

$$\begin{bmatrix} x & y & z \\ 1 & 2 & -3 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 1 & -\frac{28}{31} \end{bmatrix}$$

augmented in REF

$$\rightarrow \begin{cases} 1 \cdot x + 2 \cdot y + (-3) \cdot z = 4 \\ 0 \cdot x + 1 \cdot y + 3z = -3 \\ 0 \cdot x + 0 \cdot y + z = -\frac{28}{31} \end{cases}$$

you already solved system
of linear equations
(almost)

$$z = -\frac{28}{31} \quad \Downarrow$$

$$y + 3z = -3 \Rightarrow y + 3 \cdot \left(-\frac{28}{31}\right) = -3 \quad \Downarrow$$

$$= -\frac{9}{31}$$

$$x + 2y - 3z = 4$$

$$\Rightarrow x = -2y + 3z + 4 = -2\left(-\frac{9}{31}\right) + 3\left(-\frac{28}{31}\right) + 4 = \frac{58}{31}$$

Bottom $\rightarrow$ Top

much simpler!

$\hookrightarrow$ why bother with RREF?

$$\left[\begin{array}{cccc} 1 & 0 & 0 & \frac{58}{31} \\ 0 & 1 & 0 & -\frac{9}{31} \\ 0 & 0 & 1 & -\frac{28}{31} \end{array} \right] \rightarrow \begin{cases} x = \frac{58}{31} \\ y = -\frac{9}{31} \\ z = -\frac{28}{31} \end{cases}$$

our original augmented
matrix in RREF
(upon using E.R.O.)

$\Rightarrow$ most powerful way to solve
any system of linear equations

$$\text{SLE} \rightarrow A \xrightarrow{\text{ERO}} A_{\text{REF}} \xrightarrow{\text{ERO}_y} A_{\text{REF}} \rightarrow \text{SLE}$$

(augmented) (Almost solved)

Liology = Biology of linear Algebra

$$\begin{array}{c} x \quad y \quad z \\ \left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 3 & -1 & 1 & 5 \\ 1 & 3 & 0 & 1 \end{array} \right] \end{array}$$

Variable

in REF

C_1 : x - column

C_2 : y - column

C_3 : z - column

$$\left[\begin{array}{ccc|c} 1 & 2 & -3 & 4 \\ 0 & 1 & 3 & -3 \\ 0 & 0 & 1 & -\frac{24}{5} \end{array} \right]$$

Leading Variable

Linear equations $\rightarrow A \rightarrow A_{REF} \rightarrow A_{RREF} \rightarrow S.L.E$

REF

$\emptyset \Leftarrow$ No Solution

x	y	z	
1	0	0	2
0	1	0	7
0	0	0	1

(?) $0 \cdot x_1 + 0 \cdot x_2 + 0 \cdot x_3 = 1$
No matter what

Not leading one
in last column

$y \cdot 1 = 7$
 $x \cdot 1 = 2$

$\Rightarrow (2, 7, t)$

infinite solutions
any t works

RREF

Solve from bottom $\uparrow$

$z = t$

1	0	0	2
0	1	0	7
0	0	0	0

forget about $\uparrow$

x, y - Leading variables

z - free variable

$\hookrightarrow$ $\exists$ a solution no matter
what a free variable value

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & 2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

No bad rows

→ 1 solution or zero

Leading variables

- x_1

- x_3

- x_4

Free variables

- x_2

- x_5

∴ infinite solutions

$$\begin{bmatrix} 1 & -6 & 0 & 0 & 3 & 2 \\ 0 & 0 & 1 & 0 & 4 & 7 \\ 0 & 0 & 0 & 1 & 5 & 8 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\Rightarrow \begin{cases} x_2 = t \\ x_5 = s \end{cases}$$

Assigned parameters to all free variables

- bottom to top solve leading variables

$$x_4 + 5x_5 = 8 \Rightarrow x_4 = 8 - 5s$$



↑
s



$$x_3 + 4x_5 = 7 \Rightarrow x_3 = 7 - 4x_5 = 7 - 4s$$

$$x_1 - 6x_2 + 3x_5 = 2$$

$$x_1 - 6t + 3s = 2 \Rightarrow x_1 = 2 + 6t + 3s$$

$$(x_1, x_2, x_3, x_4, x_5)$$

$$= (2 + 6t + 3s, t, 7 - 4s, 8 - 5s, s)$$

$$\begin{cases} x_1 + 3x_2 - 2x_3 + 2x_5 = 0 \\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1 \\ 5x_3 + 10x_4 + 15x_6 = 5 \\ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6 \end{cases}$$

$b = \begin{bmatrix} 0 \\ -1 \\ 5 \\ 6 \end{bmatrix}$
 $\nearrow A \leftrightarrow$ dimension of a matrix
 $\dim(A) = 17 \times 13$
 $\uparrow \quad \quad \uparrow$
 row columns

$$\begin{aligned} \dim(\text{Augmented}) &= \\ &= \# \text{ eq} \times (\# \text{ varib} + 1) \end{aligned}$$

$$0 \cdot x_1 + 0x_2 + 5x_3 + 10x_4 + 0x_5 + 15x_6 = 5$$



$$5x_3 + 10x_4 + 15x_6 = 5$$

$$\begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{bmatrix} \equiv A$$

Practice questions on lecture 4 notes

Homogenous

$$b_1 = b_2 = b_3 \dots b_n = 0$$

All RHSs = 0

$$\begin{array}{rcl} 2x - y - 3z & = & 0 \\ -x + 2y - 3z & = & 0 \\ x + y + 4z & = & 0 \end{array}$$

Always at least one solution

$$(x, y, z) = (0, 0, 0)$$

Matrix operations

$\Rightarrow$ A matrix is a 2D array of numbers

